IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 32

Launch a Private Amazon Linux EC2 Instance with Session Manager

Amazon EC2 | Amazon Linux | Session Manager | Private Subnet | CloudFormation

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Portfolio documentation standard v1.0.1

Executive Summary

Logged into **TestAccount1**, created a **SLAW** VPC stack from a provided CloudFormation template, launched an **Amazon Linux EC2 instance**, connected to the instance using **AWS Systems Manager Session Manager**, confirmed the instance was running and reachable, then terminated the instance and deleted the CloudFormation stack to reduce cost.

SECURITY PATTERN	The important security pattern is private instance administration. The instance was accessed through Session Manager instead of SSH, avoiding a public IP, inbound SSH rule, and SSH key dependency.

Business Problem

The previous VPC and security group labs built network foundations, but no workload had been launched yet. This project establishes the first compute workload while preserving the security model: the instance runs in a private subnet, uses a managed AWS Linux image, and is accessed through Session Manager rather than exposing SSH to the Internet.

Objectives

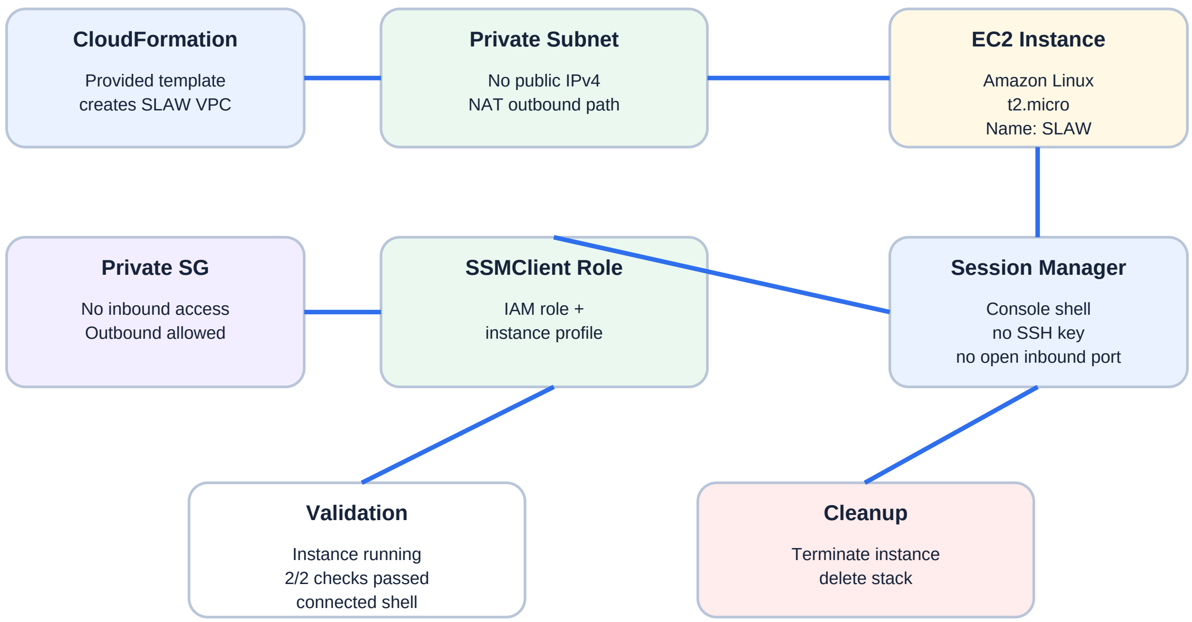
- Use the provided CloudFormation template to create the SLAW VPC and supporting access resources.
- Launch a single Amazon Linux EC2 instance.
- Use a low-cost/free-tier-friendly t2.micro instance type for the lab.
- Avoid SSH key usage and inbound SSH exposure by using Session Manager.
- Place the instance in the CloudSLAW private subnet with no public IP.
- Attach the provided private security group and SSMClient instance profile.
- Confirm the instance reaches running state and passes health checks.
- Connect to the instance using Session Manager and validate shell access.
- Terminate the instance and delete the stack for cleanup.

Environment

Component	Configuration
Cloud Provider	Amazon Web Services (AWS)
Primary Services	Amazon EC2, AWS Systems Manager Session Manager, AWS CloudFormation
Account Context	TestAccount1 / AdministratorAccess
Stack Name	SLAW
VPC	SLAW / CloudSLAW VPC created from provided template

Component	Configuration
Instance Name	SLAW
Operating System / AMI	Amazon Linux
Instance Type	t2.micro
Availability Zone	us-west-2a
Public IP / DNS	Not assigned in evidence
Connectivity	AWS Systems Manager Session Manager
IAM Access Pattern	SSMClient IAM role and instance profile from template
Security Group Pattern	Private security group with no inbound access and outbound access allowed
Cleanup	Instance terminated and CloudFormation stack deleted after validation

Architecture



Session Manager provides private instance access without SSH keys, public IPs, or inbound security group rules.

Figure 1 - Private EC2 access pattern using Session Manager.

Implementation Summary

1. Signed into TestAccount1 using AdministratorAccess.

2. Created the SLAW CloudFormation stack from the provided template.

3. Used the stack-created VPC, security group, SSMClient role, and instance profile.

4. Opened EC2 and launched a new instance named SLAW.

5. Selected Amazon Linux as the AMI.

6. Used t2.micro for cost-controlled lab testing.

7. Proceeded without an SSH key pair because Session Manager was used instead of SSH.

8. Selected the CloudSLAW VPC and a private subnet.

9. Disabled public IP assignment.

10. Selected the stack-created private security group instead of the default or launch-wizard security group.

11. Selected the SSMClient IAM instance profile.

12. Launched the instance and waited for it to enter running state and pass status checks.

13. Connected through Session Manager and validated shell access.

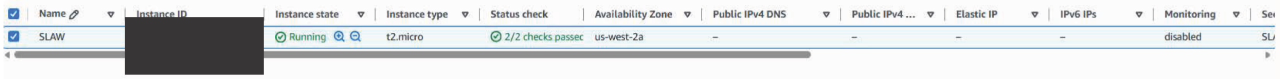
14. Terminated the instance and deleted the CloudFormation stack to reduce costs.

Control Design

Control	Configuration	Security Purpose
Private subnet	Instance launched without direct Internet inbound route	Reduces exposure to network-based Internet attacks.
No public IP	Public IPv4/DNS not assigned in evidence	Prevents direct public addressing of the instance.
No SSH key	Proceed without key pair	Avoids SSH-key handling and removes SSH as the administrative path for the lab.
Private security group	No inbound access, outbound access allowed	Allows Session Manager agent outbound communication without opening inbound ports.
SSMClient instance profile	IAM role/profile attached to EC2	Authorizes the instance to communicate with Systems Manager.
Session Manager	Console-based shell session	Provides access without bastion hosts or inbound SSH.

Evidence

Figure 2 - EC2 Instance Running and Health Checks Passed



The EC2 evidence shows the SLAW instance in Running state, type t2.micro, 2/2 status checks passed, Availability Zone us-west-2a, monitoring disabled, and no public IPv4 DNS or public IPv4 address shown. The instance ID was redacted.

Figure 3 - Session Manager Shell Connected

The Session Manager evidence shows an active browser-based terminal with a shell prompt and a Terminate session button, confirming the instance was reachable through Session Manager rather than direct SSH.

Security Analysis

This is the first compute workload in the portfolio sequence, but it follows the same security baseline built in earlier projects: private subnet placement, controlled outbound access, and identity-based administration.

Using Amazon Linux from the AWS Quick Start path reduces image-source risk compared with untrusted community AMIs, but the operating system still requires patch management in real environments.

Session Manager removes the need for a public IP, bastion host, open SSH rule, or SSH key pair for this lab.

The SSMClient instance profile is the critical authorization component. Without the IAM role and instance profile, the instance would not be able to register and communicate correctly for Session Manager access.

The private security group blocks inbound access and allows outbound traffic so the Systems Manager agent can communicate with AWS service endpoints through the network path provided by the lab VPC.

Deleting the instance and stack after validation prevents unnecessary EC2, NAT Gateway, and related infrastructure charges.

Lessons Learned

EC2 instances are AWS virtual machines based on AMIs.

AMI selection is a security decision. AWS-supported Amazon Linux AMIs are safer starting points than arbitrary community AMIs.

Session Manager can provide administrative access to private instances without inbound SSH.

Instance profile selection matters because the EC2 instance uses that role to interact with AWS services.

An instance can be running but not ready until status checks pass.

Terminating an instance is non-recoverable and deletes the associated instance resources.

Production Improvements

- Use launch templates or Auto Scaling groups instead of one-off console launches.
- Enable Session Manager session logging to CloudWatch Logs or S3.
- Use VPC endpoints for Systems Manager, EC2Messages, and SSMMessages to reduce NAT dependency.
- Patch Amazon Linux regularly with Systems Manager Patch Manager.
- Add CloudWatch alarms and AWS Config rules for instance exposure and public IP detection.
- Use IAM least privilege for the instance profile.
- Add EBS encryption, backup policy, and tagging standards.
- Restrict outbound access where possible instead of relying on broad egress.

Skills Demonstrated

Compute	Secure Administration	Cost Control
Amazon EC2	Amazon Linux	t2.micro
Session Manager	SSMClient role	Instance profile
Private subnet access	No public IP	Instance termination

Resume Bullet

RESUME	Launched a private Amazon Linux EC2 instance in a CloudFormation-created VPC, connected using AWS Systems Manager Session Manager without SSH keys or public inbound access, validated instance health and shell access, and terminated all lab resources for cost control.

STAR Interview Story

Situation: The AWS lab environment had networking and security group foundations, but no compute workload had been launched yet.

Task: Deploy a first EC2 instance while maintaining a private, secure access pattern.

Action: I created the SLAW VPC stack from the provided template, launched an Amazon Linux t2.micro instance in a private subnet, attached the SSMClient instance profile and private security group, connected through Session Manager, confirmed the instance was running, then terminated the instance and deleted the stack.

Result: The lab demonstrated private EC2 deployment and administration without public SSH exposure, while keeping the environment cost-controlled.

Version History

Version	Release Date	Description
v1.0.0	July 22, 2026	Initial documented implementation. Created SLAW VPC stack from provided template, launched Amazon Linux EC2 instance, connected with Session Manager, confirmed running status and shell access, terminated instance, deleted stack, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

• Cloud Security Lab a Week (S.L.A.W.) - Run Our First Instance (FINALLY!).

• User-provided EC2 running instance evidence screenshot.

• User-provided Session Manager connection evidence screenshot.

• Project 30 - Recreate the CloudSLAW VPC with CloudFormation IaC.

• Project 31 - Create Referenced Security Groups in a CloudFormation VPC.